

FUNDAMENTALS OF NANO AND QUANTUM PHOTONICS

Lecture 1.3: Dispersion Relation

Comprehensive Physics Study Notes & Derivations

This document serves as a self-contained, textbook-quality study guide covering the physics and mathematics of electromagnetic wave propagation and dispersion relations. Every section follows a rigorous chain of logical reasoning: starting with the fundamental physical concepts and their underlying assumptions, moving through step-by-step mathematical derivations, and concluding with deep physical interpretations.

1 The 1D Wave Equation: Foundations and Simplification

1.1 Physical Idea

Light propagates through space as a transverse electromagnetic wave containing perpendicular electric and magnetic fields that oscillate in time and space. As light transitions from a vacuum into a dielectric medium, its interaction with the medium's bound charges modifies its propagation speed and characteristics.

1.2 Fundamental Assumptions

To establish the classical wave equation, we assume:

1. The medium is a **dielectric** that is linear, isotropic, homogeneous, and non-dispersive (initially).
2. The medium is **non-magnetic**, meaning its relative permeability is unity ($\mu_r = 1$).
3. There are no free charges or currents in the medium ($\rho = 0$, $\mathbf{J} = 0$).

1.3 The 3D Wave Equation

Under these assumptions, Maxwell's equations combine to form the three-dimensional Helmholtz wave equation for the electric field vector $\mathbf{E}$:

$$\nabla^2 \mathbf{E} = \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (1)$$

where ∇^2 represents the Laplacian operator:

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad (2)$$

and μ and ϵ represent the magnetic permeability and electric permittivity of the medium, respectively.

1.4 Derivation of the 1D Wave Equation

Solving the wave equation in three dimensions is mathematically complex and can obscure core physical behavior. To simplify, we model a one-dimensional propagation case.

We assume that:

- The wave is propagating strictly along the x -axis.
- There is no transverse spatial variation of the electric field in the y and z directions ($\frac{\partial}{\partial y} = 0, \frac{\partial}{\partial z} = 0$).

This reduces the 3D Laplacian to a single second spatial derivative:

$$\frac{\partial^2 E}{\partial x^2} = \mu\epsilon \frac{\partial^2 E}{\partial t^2} \quad (3)$$

Next, we rewrite the material coefficient $\mu\epsilon$ in terms of the refractive index n and the speed of light in vacuum c . By definition:

$$\mu = \mu_r \mu_0 \quad \text{and} \quad \epsilon = \epsilon_r \epsilon_0 \quad (4)$$

where μ_0 and ϵ_0 are the permeability and permittivity of free space. The speed of light in vacuum is defined by:

$$\frac{1}{c^2} = \mu_0 \epsilon_0 \quad (5)$$

Substituting these definitions and applying our assumption of a non-magnetic medium ($\mu_r = 1$), we obtain:

$$\mu\epsilon = (1 \cdot \mu_0)(\epsilon_r \epsilon_0) = \epsilon_r (\mu_0 \epsilon_0) = \frac{\epsilon_r}{c^2} \quad (6)$$

The refractive index of a dielectric medium is defined as $n = \sqrt{\epsilon_r}$. Substituting this yields:

$$\mu\epsilon = \frac{n^2}{c^2} \quad (7)$$

Inserting this back into the 1D wave equation gives the final standard form:

$$\frac{\partial^2 E}{\partial x^2} = \frac{n^2}{c^2} \frac{\partial^2 E}{\partial t^2} \quad (8)$$

1.5 Physical Interpretation

Equation 8 is a second-order partial differential equation. Physically, it describes a dynamic coupling: the local spatial curvature of the electric field (LHS, representing how the field wraps in space) is directly proportional to its local temporal acceleration (RHS, representing how fast the field is changing in time). The proportionality constant $\frac{n^2}{c^2}$ represents the inverse square of the phase velocity in the medium ($v_p = c/n$), establishing that a denser optical medium slows down the temporal response relative to the spatial curvature.

2 The Plane-Wave Solution and Sign Conventions

2.1 Physical Idea

The most fundamental and physically insightful solution to the 1D wave equation is a plane wave. It represents a monochromatic (single-frequency) disturbance traveling infinitely through space with flat, parallel phase fronts.

2.2 Mathematical Form

We propose a complex exponential plane-wave solution propagating in the $+x$ direction:

$$E(x, t) = E_0 e^{i(kx - \omega t)} \quad (9)$$

2.3 Symbol Definitions

- E_0 : The complex wave amplitude, containing both the peak physical amplitude and any initial phase offset.
- e : The base of the natural logarithm.
- i : The imaginary unit ($i^2 = -1$).
- k : The wave vector (or wavenumber in 1D), measured in radians per meter (rad/m). It represents the spatial frequency of the wave.
- ω : The angular frequency, measured in radians per second (rad/s). It represents the temporal oscillation rate.
- x : Spatial coordinate along the propagation direction (m).
- t : Time coordinate (s).
- $\phi(x, t) = kx - \omega t$: The instantaneous phase of the wave (radians).

2.4 Physical Meaning and Wavelength Connections

1. **Why it is a "Plane" Wave:** A wave is designated a "plane wave" because its surfaces of constant phase form flat, parallel planes perpendicular to the direction of propagation. Mathematically, setting the instantaneous phase $\phi(x, t)$ to a constant value at a fixed time t_0 yields:

$$kx - \omega t_0 = \text{constant} \implies x = \text{constant} \quad (10)$$

The equation $x = \text{constant}$ defines a flat 2D plane spanning the transverse y - and z -axes. Since the wave has no transverse spatial dependence (y or z), every point on this plane has the exact same electric field value and phase at any instant.

2. **Wavelength and Wave Vector:** The wave vector k dictates the spatial periodicity of the wave. The distance over which the phase repeats by 2π is the wavelength λ in the medium:

$$k \cdot \lambda = 2\pi \implies \lambda = \frac{2\pi}{k} \quad (11)$$

3. **Frequency and Period:** The angular frequency ω dictates the temporal periodicity. The time over which the phase repeats by 2π is the period T :

$$\omega \cdot T = 2\pi \implies T = \frac{2\pi}{\omega} \implies \omega = 2\pi f \quad (12)$$

where f is the cyclic frequency in Hertz (Hz).

4. **Propagation Direction:** The sign combination in the exponent determines the propagation direction. For $\phi = kx - \omega t$, as time t increases, a point of constant phase must move toward larger x to keep $kx - \omega t$ constant. Thus, the wave travels in the $+x$ direction. Conversely, a solution of the form $e^{i(-kx - \omega t)} = e^{-i(kx + \omega t)}$ propagates in the $-x$ direction.

2.5 Sign Conventions: Physics vs. Engineering

The wave equation also allows a second, mathematically equivalent class of solutions:

$$E(x, t) = E_0 e^{-i(kx - \omega t)} = E_0 e^{-ikx + i\omega t} \quad (13)$$

The choice of time-dependence sign defines a major notation convention:

- **Physics Convention** ($e^{-i\omega t}$): Uses a minus sign for the time term, writing the forward wave as $e^{i(kx-\omega t)}$. This is the standard adopted in physics, quantum mechanics, and this course.
- **Engineering Convention** ($e^{+i\omega t}$): Uses a plus sign for the time term, writing the forward wave as $e^{i(\omega t-kx)}$ or $e^{-i(kx-\omega t)}$. This is standard in electrical engineering and RF wave theory.

This is a purely formal notation choice that does not alter any physical results. However, it requires a careful, self-consistent definition of other physical parameters (such as the complex refractive index) to ensure that passive materials absorb energy rather than physically amplifying it.

3 Energy Propagation and the Poynting Vector

3.1 Physical Idea

Electromagnetic waves are not merely abstract mathematical solutions; they transport electromagnetic energy through space.

3.2 Derivation and Orthogonal Fields

From Maxwell's equations (specifically Faraday's and Ampère's laws), the electric and magnetic fields are intrinsically coupled. For a forward-propagating plane wave:

- The electric field $\mathbf{E}$ and magnetic field $\mathbf{H}$ are perpendicular to one another.
- Both fields are strictly perpendicular to the direction of wave propagation ($\hat{\mathbf{k}}$).

If the electric field is polarized along the z -axis ($\mathbf{E} = E_z \hat{\mathbf{z}}$) and propagates along the x -axis ($\hat{\mathbf{k}} = \hat{\mathbf{x}}$), the magnetic field is naturally forced into the y -axis ($\mathbf{H} = H_y \hat{\mathbf{y}}$).

3.3 The Poynting Vector

The spatial direction and instantaneous rate of this energy transport per unit area is defined by the Poynting vector $\mathbf{S}$:

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (14)$$

Since optical fields oscillate at frequencies on the order of 10^{14} Hz, macroscopic detectors cannot measure the instantaneous energy fluctuations. Instead, they record the time-averaged energy flux, given by the time-averaged Poynting vector:

$$\langle \mathbf{S} \rangle = \frac{1}{2} \text{Re} \{ \mathbf{E} \times \mathbf{H}^* \} \quad (15)$$

The speed at which this time-averaged energy packet travels through space is governed by the group velocity of the medium.

4 Step-by-Step Derivation of the Dispersion Relation

4.1 Physical Idea

The dispersion relation is the core equation that couples a wave's spatial frequency (k) and temporal frequency (ω) inside a given medium of refractive index n .

4.2 Assumptions

We assume that the plane-wave solution $E(x, t) = E_0 e^{i(kx - \omega t)}$ must satisfy the 1D wave equation in our medium: $\frac{\partial^2 E}{\partial x^2} = \frac{n^2}{c^2} \frac{\partial^2 E}{\partial t^2}$.

4.3 Step-by-Step Mathematical Derivation

We begin by taking the spatial and temporal derivatives of our proposed plane-wave solution:

$$E(x, t) = E_0 e^{i(kx - \omega t)} \quad (16)$$

Step 1: Compute the first spatial derivative Using the chain rule:

$$\frac{\partial E}{\partial x} = \frac{\partial}{\partial x} [E_0 e^{i(kx - \omega t)}] = (ik) \cdot E_0 e^{i(kx - \omega t)} = ikE(x, t) \quad (17)$$

Step 2: Compute the second spatial derivative (LHS of wave equation) Differentiating again with respect to x :

$$\frac{\partial^2 E}{\partial x^2} = \frac{\partial}{\partial x} [ikE(x, t)] = (ik)(ik)E(x, t) = i^2 k^2 E(x, t) \quad (18)$$

Since $i^2 = -1$, the second spatial derivative is:

$$\frac{\partial^2 E}{\partial x^2} = -k^2 E(x, t) \quad (19)$$

Step 3: Compute the first temporal derivative Using the chain rule with respect to t :

$$\frac{\partial E}{\partial t} = \frac{\partial}{\partial t} [E_0 e^{i(kx - \omega t)}] = (-i\omega) \cdot E_0 e^{i(kx - \omega t)} = -i\omega E(x, t) \quad (20)$$

Step 4: Compute the second temporal derivative (RHS of wave equation) Differentiating again with respect to t :

$$\frac{\partial^2 E}{\partial t^2} = \frac{\partial}{\partial t} [-i\omega E(x, t)] = (-i\omega)(-i\omega)E(x, t) = i^2 \omega^2 E(x, t) \quad (21)$$

Since $i^2 = -1$, the second temporal derivative is:

$$\frac{\partial^2 E}{\partial t^2} = -\omega^2 E(x, t) \quad (22)$$

Step 5: Substitute derivatives into the 1D Wave Equation Substitute Eq. 19 and Eq. 22 into Eq. 8:

$$-k^2 E(x, t) = \frac{n^2}{c^2} (-\omega^2 E(x, t)) \quad (23)$$

Assuming a non-trivial wave solution ($E(x, t) \neq 0$), we can divide both sides by $-E(x, t)$:

$$k^2 = \frac{n^2}{c^2} \omega^2 \quad (24)$$

Taking the square root of both sides yields the dispersion relation:

$$\omega = \pm \frac{c}{n} k \quad (25)$$

4.4 Physical Meaning

Equation 25 establishes that in any medium, the wave vector k and angular frequency ω are not independent parameters; they are locked in a strict proportional relationship dictated by the material's refractive index n and the universal speed of light c . The plus-or-minus sign represents waves traveling in the positive ($+x$) and negative ($-x$) spatial directions, respectively.

5 Visualizing and Reading the Dispersion Graph ($\omega - k$)

5.1 Vacuum Dispersion ($n = 1$)

In free space (vacuum), the refractive index $n = 1$. The dispersion relation reduces to:

$$\omega = \pm ck \quad (26)$$

This is a perfectly linear relationship.

- **Graphical Representation:** In a 1D plot with k on the horizontal axis and ω on the vertical axis, this plots as two straight lines passing through the origin with a slope of $\pm c$. In three dimensions (k_x, k_y, ω), this linear relationship sweeps out a circular cone known as the **light cone**.

5.2 Lossless Medium Dispersion ($n > 1$)

In a simple, transparent dielectric material (like glass), the refractive index $n > 1$. The dispersion relation becomes:

$$\omega = \pm \frac{c}{n} k \quad (27)$$

Because $n > 1$, the slope of the $\omega - k$ line ($\pm c/n$) is **reduced** compared to the vacuum line.

- **Graphical Representation:** The dispersion line in a medium lies at a shallower angle relative to the k -axis than the vacuum line. The higher the refractive index, the lower the slope of the dispersion curve.

5.3 How to Read the Graph: The Frequency Invariance Principle

A major source of confusion in photonics is reading the $\omega - k$ graph as if varying the wave vector k changes the frequency ω . This is physically backward.

The Source Rule

The angular frequency ω of an electromagnetic wave is a **fundamental property of its source**. Once a wave is generated (for example, by an atomic transition or a laser oscillating at frequency ω_0), this frequency **cannot change** when the wave enters a different medium. The time-varying oscillation of the field is locked at the boundary.

To read the dispersion relation correctly:

1. Identify the fixed operating frequency ω_0 on the vertical axis.
2. Project a horizontal line across the graph at $\omega = \omega_0$.
3. Find the intersection of this line with the dispersion curves of the different media:
 - In vacuum ($n = 1$), the intersection yields the free-space wave vector:

$$k_0 = \frac{\omega_0}{c} \quad (28)$$

- In a dielectric medium ($n > 1$), the intersection yields a larger wave vector:

$$k = n \frac{\omega_0}{c} = nk_0 \quad (29)$$

Since $k = 2\pi/\lambda$, the physical consequence of k increasing in the medium is that the wavelength must shrink:

$$\lambda_{\text{medium}} = \frac{\lambda_0}{n} \quad (30)$$

Thus, when light enters a medium, **frequency remains perfectly constant**, while the wave vector increases and the wavelength compresses by a factor of n .

6 Non-Linear Dispersion and Guided Modes

In bulk lossless dielectrics, the dispersion relation is linear over a broad frequency range. However, this is not a universal rule.

6.1 Physical Idea

When light is confined inside a micro-structure, such as a waveguide or optical fiber, the boundary conditions restrict the allowed solutions of the wave equation to discrete spatial patterns called **guided modes**.

6.2 Non-Linear Dispersion Curves

The confinement introduces a geometric dispersion that modifies the relationship between ω and k , making the dispersion relation highly **non-linear**.

- Guided modes can exhibit quadratic, parabolic, or complex shapes on the $\omega - k$ plot.
- In some waveguiding structures, the dispersion curve can become extremely flat ($\frac{d\omega}{dk} \approx 0$) in specific frequency bands. This represents a regime where the group velocity of light drops toward zero, enabling the engineering of "slow light" devices.

7 Velocities of Wave Propagation: Phase vs. Group Velocity

When dispersion becomes non-linear, the concept of a single "speed of light" in a medium breaks down. We must distinguish between two separate velocities.

7.1 Phase Velocity (v_p)

The phase velocity is the speed at which a single, monochromatic wave's point of constant phase (such as a single crest or trough) travels through space.

$$v_p = \frac{\omega}{k} \quad (31)$$

Using the standard dispersion relation $\omega = \frac{c}{n}k$, the phase velocity in a medium is:

$$v_p = \frac{c}{n} \quad (32)$$

For a pure, infinite sinusoidal wave, no information or energy can be sent by the wave itself because it has no beginning, end, or modulation; its individual phase fronts simply glide forward at v_p .

7.2 Group Velocity (v_g)

The group velocity is the speed at which the envelope of a wave packet (a localized group of waves containing a narrow band of frequencies) or the wave's energy propagates through space. It is mathematically defined as the first derivative of the dispersion relation:

$$v_g = \frac{d\omega}{dk} \quad (33)$$

7.3 Physical Distinctions and Energy Flow

- **Phase Velocity ($v_p = \omega/k$):** Represents the speed of phase propagation. In some artificial systems (such as plasmonic structures or waveguides near cutoff), v_p can exceed the speed of light in vacuum c , which does not violate relativity because phase fronts carry no energy or information.
- **Group Velocity ($v_g = d\omega/dk$):** Represents the speed of energy and information propagation. It is the velocity of the signal's wave packet. According to special relativity, v_g for any real physical signal cannot exceed c in a passive medium.
- **Linear vs. Non-linear Mediums:**
 - In a perfectly linear, non-dispersive medium, $\omega \propto k$, making the slope $\frac{d\omega}{dk}$ equal to the ratio $\frac{\omega}{k}$. Thus, $v_g = v_p = c/n$.
 - In a dispersive or waveguiding medium, the dispersion curve is non-linear, meaning the derivative (slope) $\frac{d\omega}{dk}$ varies with frequency and differs from the absolute ratio $\frac{\omega}{k}$. Consequently, $v_g \neq v_p$.

8 Complex Refractive Index in Absorbing Media

8.1 Physical Idea

In lossless, transparent materials, the refractive index is modeled as a purely real number. However, no real-world material is perfectly transparent; some portion of the propagating wave's electromagnetic energy is always absorbed by the medium (e.g., via coupling to electronic transitions or phonon vibrations). To capture this absorption mathematically, we must transition to a complex refractive index.

8.2 Mathematical Representation

We define the complex refractive index n as:

$$n = n' + in'' \quad (34)$$

where:

- n' is the real part of the refractive index, representing the refractive phase velocity scale.
- n'' is the imaginary part of the refractive index, representing the absorption parameter of the medium.

8.3 Step-by-Step Derivation of Wave Attenuation

We now trace how this complex index affects the wave mathematically when inserted back into our 1D plane wave solution.

Step 1: Calculate the complex wave vector k According to the dispersion relation, the wave vector k is given by:

$$k = \frac{\omega}{c}n \quad (35)$$

Substituting $n = n' + in''$ yields:

$$k = \frac{\omega}{c}(n' + in'') = \frac{\omega n'}{c} + i\frac{\omega n''}{c} \quad (36)$$

Thus, the wave vector k itself becomes a complex quantity: $k = k' + ik''$, where the real part is $k' = \frac{\omega n'}{c}$ and the imaginary part is $k'' = \frac{\omega n''}{c}$.

Step 2: Substitute k back into the 1D plane-wave solution The forward plane wave is written as:

$$E(x, t) = E_0 e^{i(kx - \omega t)} \quad (37)$$

Substituting our complex k (Eq. 36) into the spatial term:

$$E(x, t) = E_0 e^{i\left[\left(\frac{\omega n'}{c} + i\frac{\omega n''}{c}\right)x - \omega t\right]} \quad (38)$$

Step 3: Expand and separate exponents Distribute the term i across the spatial components in the exponent:

$$E(x, t) = E_0 e^{i\frac{\omega n'}{c}x + i^2\frac{\omega n''}{c}x - i\omega t} \quad (39)$$

Since $i^2 = -1$:

$$E(x, t) = E_0 e^{i\frac{\omega n'}{c}x - \frac{\omega n''}{c}x - i\omega t} \quad (40)$$

Group the oscillatory and decaying components into separate exponential terms:

$$E(x, t) = E_0 e^{i\left(\frac{\omega n'}{c}x - \omega t\right)} e^{-\frac{\omega n''}{c}x} \quad (41)$$

8.4 Physical Interpretation of the Two Terms

Equation 41 splits the propagating electromagnetic field into two distinct, multiplicative physical terms:

1. **The Propagating Wave Term:** $e^{i\left(\frac{\omega n'}{c}x - \omega t\right)}$ This is a complex exponential phase factor. As position x increases, this term oscillates sinusoidally without decay. This term represents propagation. Its phase velocity is determined by:

$$v_p = \frac{\omega}{k'} = \frac{c}{n'} \quad (42)$$

Thus, the real part n' controls the phase velocity and refraction of the wave.

2. **The Attenuation Term:** $e^{-\frac{\omega n''}{c}x}$ This is a real-valued decaying exponential function. As the wave travels along the x -axis, its amplitude decays exponentially. The decay rate is proportional to n'' . Thus, the imaginary part n'' produces absorption/attenuation of the wave. If $n'' = 0$, there is no decay, and the material is transparent.

9 Amplitude vs. Intensity Attenuation

In electromagnetism, we must distinguish between the attenuation of the field amplitude and the attenuation of the wave's power or intensity.

9.1 The Squaring Factor of 2

The physical intensity $I(x)$ of an electromagnetic wave represents the optical power per unit area, which is proportional to the time-averaged Poynting vector, and hence is proportional to the square of the electric field's physical amplitude:

$$I(x) \propto |E(x, t)|^2 = E(x, t) \cdot E^*(x, t) \quad (43)$$

where $E^*(x, t)$ is the complex conjugate of the electric field. Substituting our attenuated wave solution (Eq. 41):

$$|E(x, t)|^2 = \left[E_0 e^{i\left(\frac{\omega n'}{c}x - \omega t\right)} e^{-\frac{\omega n''}{c}x} \right] \cdot \left[E_0^* e^{-i\left(\frac{\omega n'}{c}x - \omega t\right)} e^{-\frac{\omega n''}{c}x} \right] \quad (44)$$

$$= |E_0|^2 \cdot e^{i(0)} \cdot e^{-2\frac{\omega n''}{c}x} \quad (45)$$

$$= |E_0|^2 e^{-2\frac{\omega n''}{c}x} \quad (46)$$

Therefore, the physical intensity of the light decays according to the Beer-Lambert law:

$$I(x) = I_0 e^{-2\frac{\omega n''}{c}x} = I_0 e^{-\alpha x} \quad (47)$$

where I_0 is the initial intensity at $x = 0$, and α is the physical **intensity absorption coefficient**:

$$\alpha = 2\frac{\omega}{c}n'' \quad (48)$$

9.2 Physical Meaning

The factor of 2 in the exponent of the intensity decay shows that **intensity decays twice as fast as the electric field amplitude**. This occurs because intensity is a quadratic (square) function of the electric field. When defining material decay parameters, we distinguish:

- **Propagation Constant (β):** Associated with the real phase propagation: $\beta = k' = \frac{\omega}{c}n'$.
- **Attenuation Constant (Field Decurrence):** Associated with the decay of the field amplitude: $\frac{\omega}{c}n''$.
- **Absorption Coefficient (α):** Associated with the decay of physical intensity/power: $\alpha = 2\frac{\omega}{c}n''$.

10 Notation vs. Physics: The Consistency of Sign Conventions

As discussed in Section 2, choosing a time-dependence sign is a convention, but it creates a strict mathematical dependency when defining the complex refractive index.

If we use the **Physics Convention** ($e^{-i\omega t}$), our plane wave is $e^{i(kx - \omega t)}$:

- We **must** define $n = n' + in''$ with a **plus sign**.
- This yields a complex wave vector $k = \frac{\omega n'}{c} + i\frac{\omega n''}{c}$.
- When substituted, the spatial phase term becomes:

$$e^{i(kx)} = e^{i\left(\frac{\omega n'}{c} + i\frac{\omega n''}{c}\right)x} = e^{i\frac{\omega n'}{c}x} e^{-\frac{\omega n''}{c}x} \quad (49)$$

which yields a physical, decaying wave.

If we use the **Engineering Convention** ($e^{+i\omega t}$), our forward plane wave is written as $e^{-i(kx-\omega t)}$:

- We **must** define $n = n' - in''$ with a **minus sign**.
- This yields a complex wave vector $k = \frac{\omega n'}{c} - i\frac{\omega n''}{c}$.
- When substituted into the engineering exponent:

$$e^{-i(kx)} = e^{-i\left(\frac{\omega n'}{c} - i\frac{\omega n''}{c}\right)x} = e^{-i\frac{\omega n'}{c}x} e^{-\frac{\omega n''}{c}x} \quad (50)$$

which also yields a physical, decaying wave.

10.1 The Hazard of Inconsistency

If a physicist using the $e^{-i\omega t}$ time-dependence notation incorrectly adopts the engineering index convention $n = n' - in''$:

$$k = \frac{\omega}{c}(n' - in'') \implies e^{ikx} = e^{i\frac{\omega n'}{c}x} e^{+\frac{\omega n''}{c}x} \quad (51)$$

The attenuation term becomes $e^{+\frac{\omega n''}{c}x}$. This represents a wave that **grows exponentially** as it propagates forward, indicating that a passive, unpumped piece of glass is generating infinite optical energy. This is a violation of the conservation of energy and is physically impossible. This highlights that sign conventions are not "just notation"—they require strict mathematical consistency to preserve physical reality.

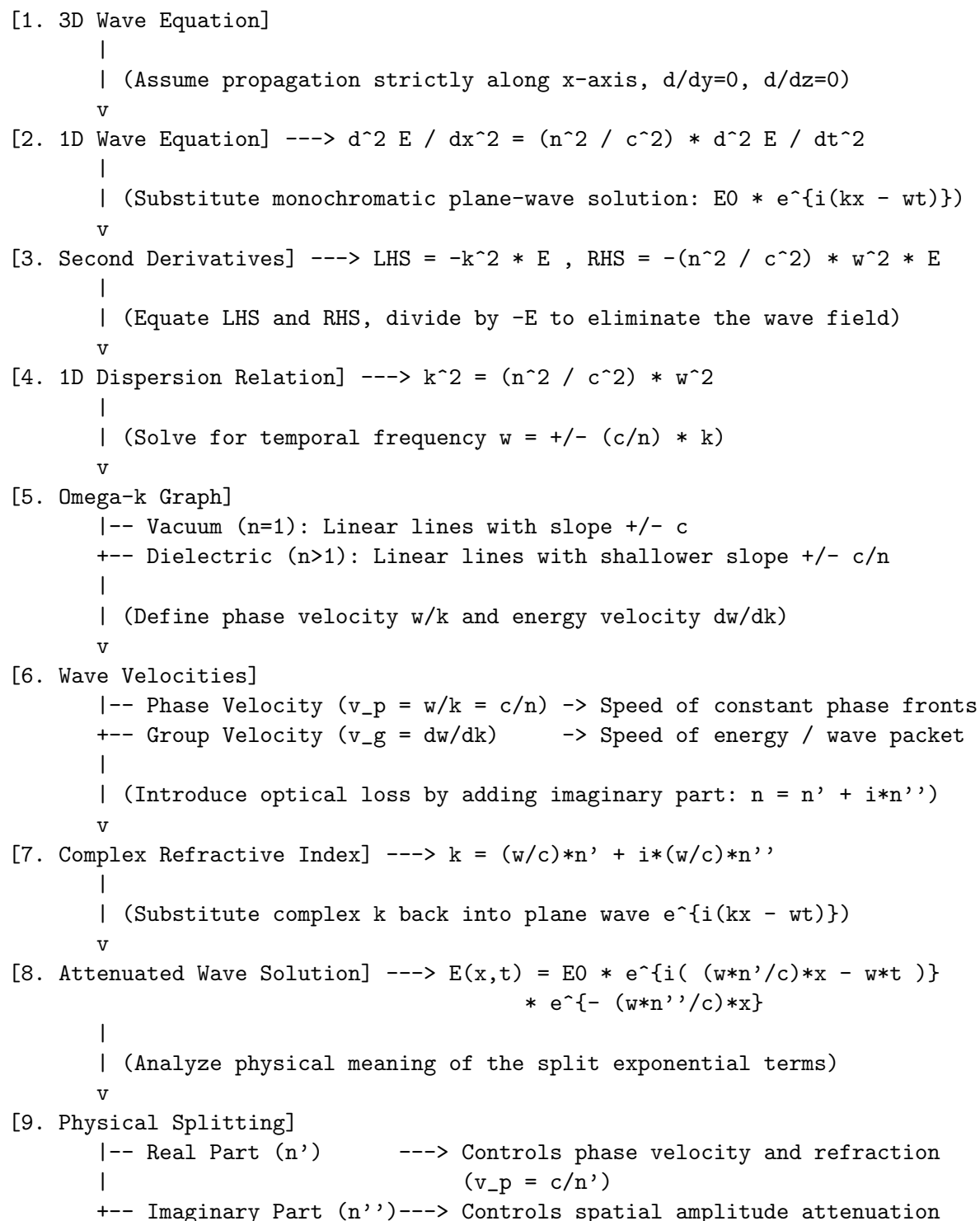
11 Common Confusions in Dispersion Theory

- **Frequency vs. Wavelength:** Frequency is a temporal parameter determined solely by the source and is invariant upon changing materials. Wavelength is a spatial parameter that compresses in an optical medium ($\lambda = \lambda_0/n$) because the wave fronts must crowd together to match the slower phase velocity.
- **Wave Vector k vs. Angular Frequency ω :** ω is temporal oscillation frequency (rad/s), while k is spatial oscillation frequency (rad/m). The dispersion relation is the material-dependent mapping between them.
- **Phase Velocity vs. Group Velocity:** Phase velocity ($v_p = \omega/k$) is the speed of individual phase fronts, which can exceed c without violating relativity. Group velocity ($v_g = d\omega/dk$) is the speed of the wave packet envelope and energy propagation, which is strictly limited to $\leq c$ in passive media.
- **Real vs. Imaginary Refractive Index (n' vs. n''):** The real part n' governs phase accumulation, refraction, and phase velocity ($v_p = c/n'$). The imaginary part n'' governs optical loss/attenuation. They are not independent; they are linked by causality.
- **Amplitude vs. Intensity Attenuation:** The electric field amplitude decays as $e^{-\gamma x}$ where $\gamma = \frac{\omega n''}{c}$. The physical power (intensity) decays as $e^{-2\gamma x}$ because intensity is proportional to the square of the field. Thus, the intensity absorption coefficient α is twice the amplitude decay rate.
- **Sign Conventions ($e^{-i\omega t}$ vs. $e^{+i\omega t}$):** Choosing the sign of time-dependence is arbitrary, but it locks the sign of the complex refractive index. Physics ($e^{-i\omega t}$) requires $n' + in''$; Engineering ($e^{+i\omega t}$) requires $n' - in''$. Swapping them incorrectly results in unphysical exponential gain.

- **Linear vs. Non-linear Dispersion:** Linear dispersion ($\omega \propto k$) occurs in simple bulk media and implies that all frequencies travel at the same velocity ($v_g = v_p$). Non-linear dispersion occurs in waveguides or near material resonances, resulting in wave-packet spreading (dispersion) and a divergence between v_g and v_p .

12 The Complete Conceptual Chain

The following flow chart map summarizes the entire logical architecture of dispersion relation theory as presented in the lecture:



$$e^{\{-(w*n''/c)*x\}}$$

|

| (Square the electric field to calculate the measurable intensity I
| proportional to $|E|^2$)

v

[10. Beer-Lambert Loss] ---> $I(x) = I_0 * e^{\{-2*(w*n''/c)*x\}} = I_0 * e^{\{-\alpha*x\}}$

|

v

[11. Final Physical State] ---> Intensity decays twice as fast as field amplitude
($\alpha = 2*w*n''/c$)